

SINGLETON DESIGN PATTERN

- ① DEFINITION
- ② V1 - Lazy Initialization
- ③ V2 - Eager Initialization
- ④ V3 - Thread-Safe
- ⑤ V4 - Double Checked Locking
- ⑥ V5 - Enum

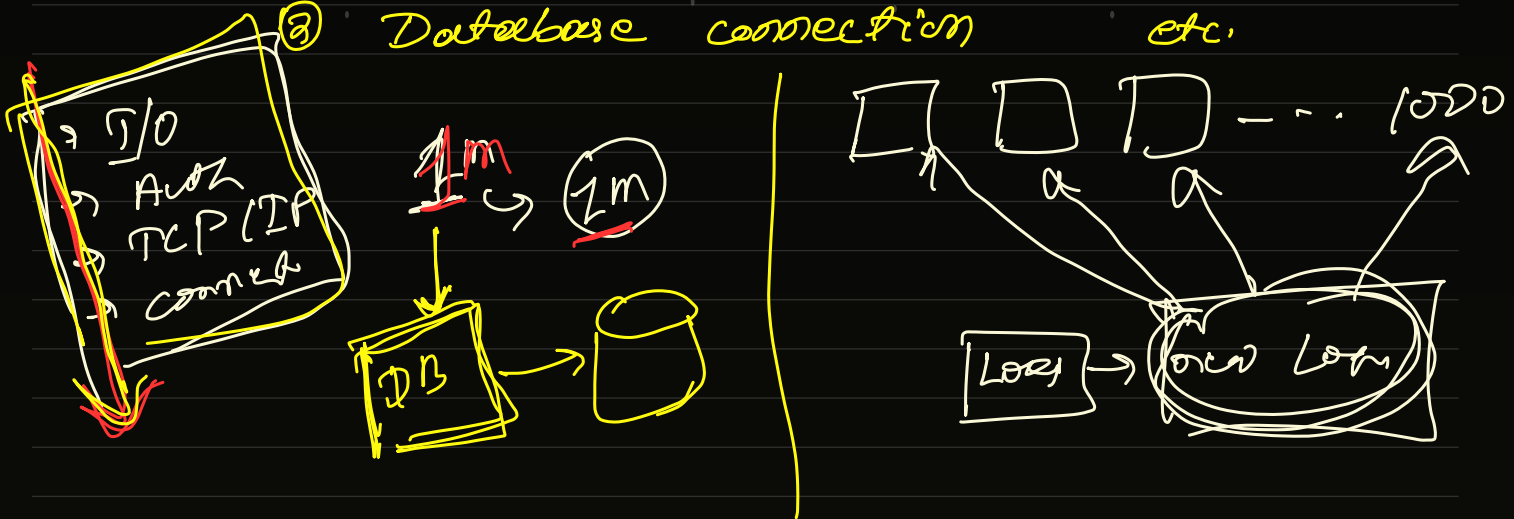
Def. [Comes under creational pattern]

We can have only one object for whole application.

Example

⊕ Logging System

② Database connection etc.



Benefits

- ① Memory Efficiency :- It saves memory by not creating multiple identical objects.
- ② Data Consistency :- Since everyone uses the same instance, there is no risk of different parts of app having different version of data.

How to create Singleton class (java)

- ① Private Constructor
- ② Private static instance
- ③ Public getter method
- ④ Lazy/Eager initialization